STUDEXPO-2026
Mobile apps and artificial intelligence

AgroZen

An AI mentor in your smartphone for beginner poultry farmers

Working MVP · Android | **3 languages** | **Works offline**

Smart barn sensors | **Reports for banks**



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2017

Families got loans to raise chickens. Reportedly, many couldn't repay, and banks cut such loans.

From 1 August 2026

Concessional poultry loans are back: up to 10 years at 10% a year.

THE PROBLEM

Loan taken, flock lost

No knowledge

A beginner doesn't know the norms for temperature, feeding and vaccination.

The first 14 days are the most dangerous

One microclimate mistake leads to mass mortality — and nobody measures ammonia or CO2.

The specialist is far away

There is no vet nearby at the moment a question comes up.

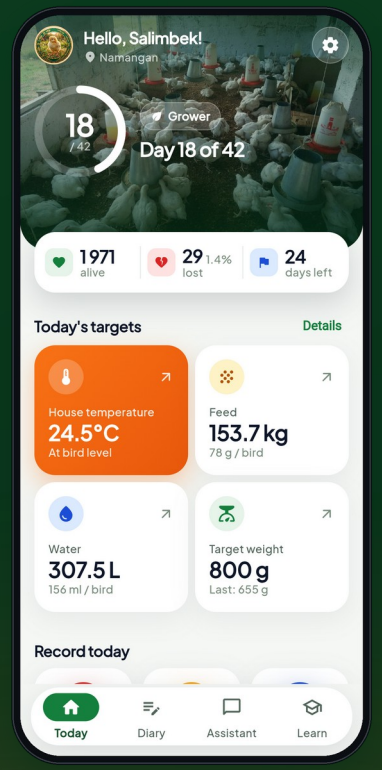


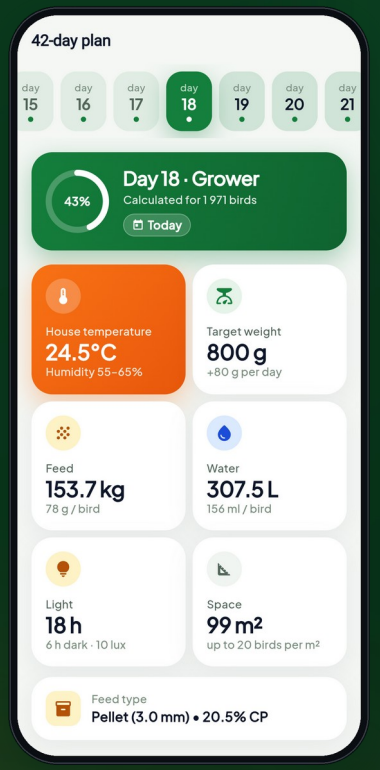
The birds are gone. The debt remains.

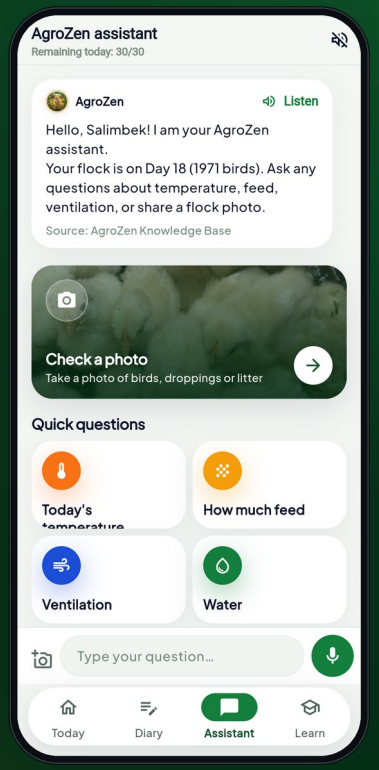
People lose money not because of laziness, but because of mistakes no one warned them about.

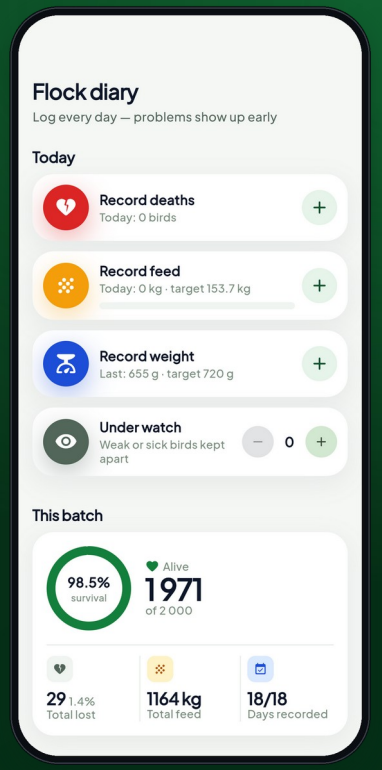
THE SOLUTION

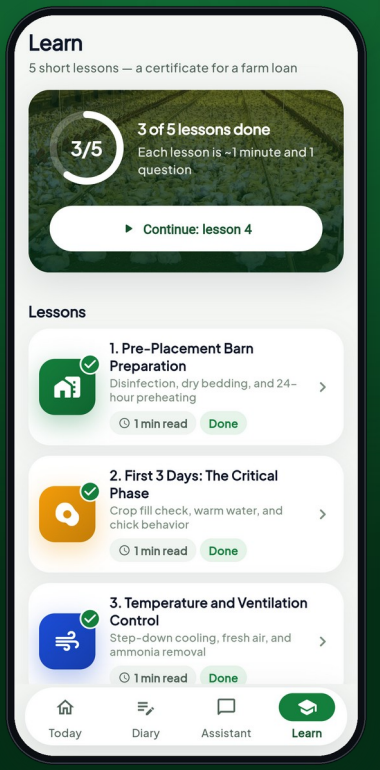
AgroZen guides the farmer through all 42 days











Today
Targets for temperature, feed and water

42-day plan
Ross-308 / Cobb-500 for every day

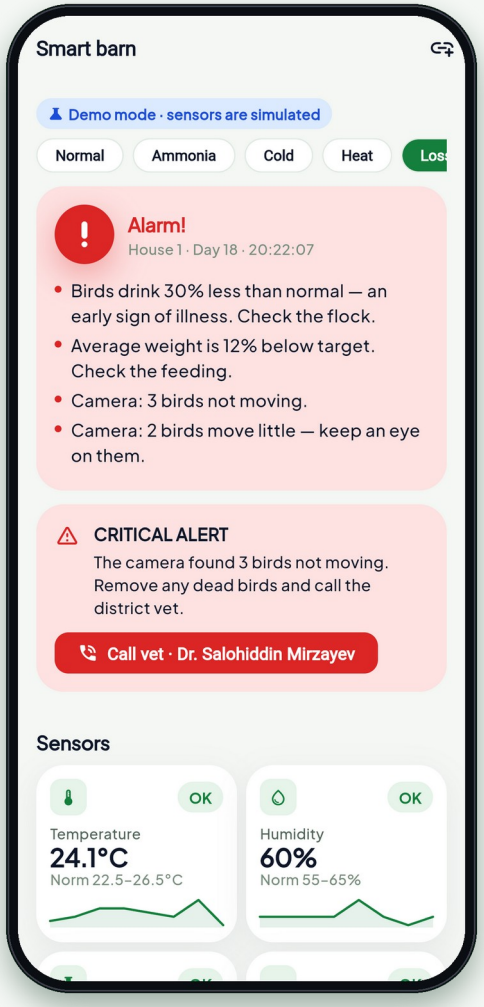
AI assistant
Voice, text or photo, in your language

Flock diary
Losses above the norm: the vet's number

Course
5 lessons and a certificate for a loan

NEW · SMART BARN PROTOTYPE

The AI camera colours every bird by its health



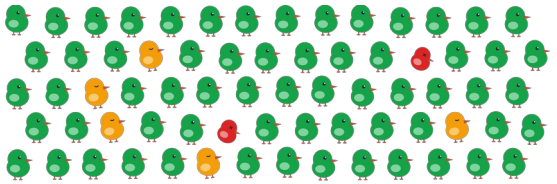


Active
Moves, eats and drinks

Moving little
An early sign: watch it

Not moving
Remove it, call the vet

Flock at a glance



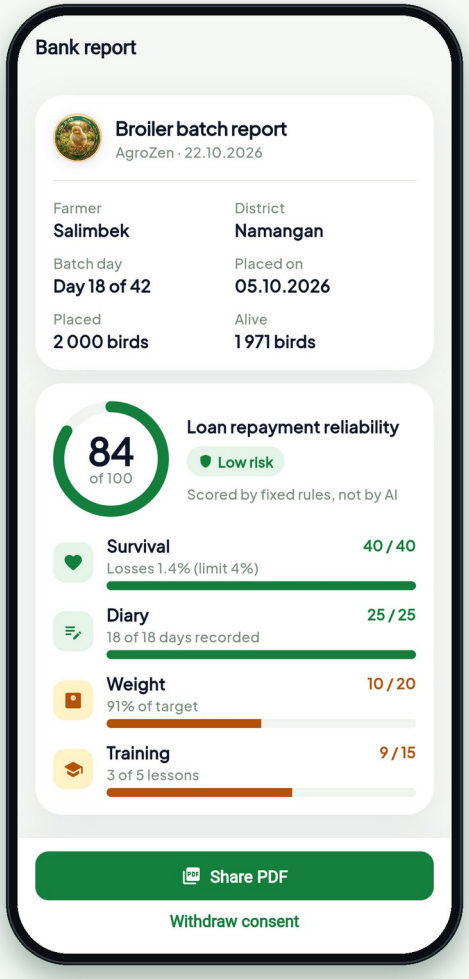
63 active · 5 watch · 2 not moving

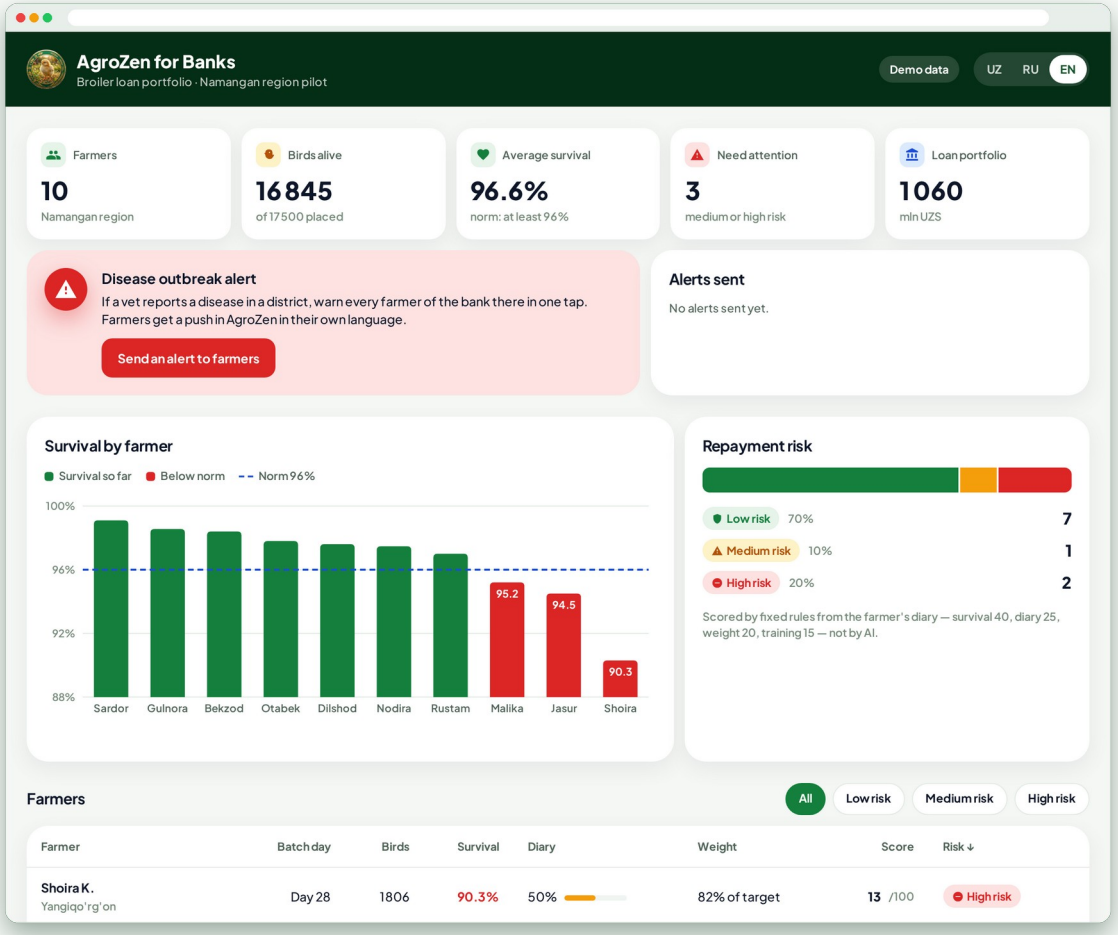
ESP32 sensors · **Jetson edge AI** · **Ross-308 rules** · **Alert · vet · fans**

NH3 up to 10 ppm · CO2 up to 3 000 ppm · fans switch on by themselves. Demo mode today; hardware goes into two pilot farms in 2027.

NEW · FOR BANKS

Banks see how the flock is doing — and trust the loan





Bank dashboard: 10 pilot farmers, survival against the norm, risk and a district disease alert

Farmer's consent
Built from the diary, sent as a PDF only after the farmer agrees.

Score 0–100
Survival, diary, weight, training: fixed rules, not AI.

Disease alert
One tap warns every farmer of the bank in a district.

Free for farmers. Banks and microfinance pay: a trained farmer is more likely to repay.

THE RESULT

Not a concept: the app already works

42 days
of norms, one plan for every day

5 lessons
a course with tests and a certificate

13 regions
vet contacts across Uzbekistan

260 tests
check norms, alerts and scores

Flutter · Supabase + Gemini · Works offline · uz · ru · en

Alerts are counted by plain code against the norms — they never depend on an AI answer.



AgroZen assistant
Answers only from a verified knowledge base

My chicks are crowding under the heater. What should I do?

voice · text · photo

That is a sign of cold. On day 3 the house should be 31.5°C — raise the heating and check that the chicks spread evenly.
Source: AgroZen knowledge base · Ross-308

Not in the base? "Call your district vet."
No diagnoses, no medicines, no dosages — ever.

NEXT STEP

A pilot with 10 farmers and a bank in Namangan region

92–95%
flock survival

80%+
days with diary entries

96%+
AI answers matching the norms

Target figures per flock (42 days).

- Sensors and AI cameras in two pilot farms
- Bank dashboard with real pilot data
- Then laying hens, sheep and cattle, greenhouses



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Download and try it
Android app
zedrubilan.github.io/agrozen



Thank you!